

DW and EDA Final Project Report

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Due: 4/4/2026

Introduction and Motivation

Popular online video sharing platforms can generate a large amount of interesting and useable data. Particularly, YouTube being the most popular of these platforms has billions of videos and hundreds of millions of daily users. YouTube has a feature called *trending charts* where typically a limited number of videos can be shown. Videos on the trending charts provide a “dynamic daily spotlight on the most popular content from across the globe” (Google, *Trending charts on YouTube*). These trending charts aim to show you the fastest growing content on YouTube, or in other words the hottest content right now. YouTube takes into consideration the view count, how quickly the view count grows, the topic, where the views come from, and the age of the video. Due to this, YouTube’s trending charts are a great source of data for exploring trends among different times and countries, studying engagement, and overall patterns of viral videos. Therefore, in this paper we propose a project where we use the dataset “YouTube Trending Videos Stats 2026” by Shul Ilya to explore trending YouTube videos. We plan on using both traditional and large language model (LLM) assisted methods of data wrangling (DW) and exploratory data analysis (EDA) to complete our work.

Related Work

We initially reviewed multiple sources dealing with the integration of LLMs into DW and EDA processes. The main sources we used before our implementation of LLMs include “Can LLMs Clean Up Your Mess? A Survey of Application-Ready Data Preparation with LLMs”, “Can language models automate data wrangling?”, and “Large Language Models as Data Preprocessors”. Each source looks into using LLMs for tasks like data cleaning, data integration, or data enrichment. The sources tend to point out that while LLMs have strong benefits in these areas, they also have significant limitations such as heavy computational load and the ability to hallucinate. While computational load as an issue is situational, hallucinations can be detrimental to any project, especially when fully automating tasks. There would need to be rigorous validation of the LLMs' work to ensure correctness and completeness. This led us to our decision to avoid full automation using LLMs. We also reviewed “ellmer: chat with large language models” for our integration approach. In addition to this, we reviewed the study “Trending Videos: Measurement and Analysis”, which analyzes the lifecycle of trending videos. Outside of this, “R for Data Science” was used as a guide that provides foundational techniques for using R for DW and EDA tasks.

Dataset Description and Source

For our project, we plan to use the “YouTube Trending Videos Stats 2026” dataset by Shul Ilya found on Kaggle. This dataset has 178,400 records and 13 variables. It includes trending YouTube videos from 11 countries, including Brazil, Canada, Germany, France, the United Kingdom, India, Japan, South Korea, Mexico, Russia, and the United States. There are 11 total files, each being for one of the 11 countries. The videos published go

back as far as 2007. The variables of this dataset include a unique video ID, the trending date, the title of the video, the channels display title, the view count, the like count, the dislike value, the published timestamp, the YouTube category ID, the video tags, comments count, YouTube channel ID, and video description. This data was collected by the authors using the YouTube Data API v3.

Preliminary Data Quality Assessment

For the most part, outside of two specifically major issues, this dataset is high quality. First, the data is collected using the official YouTube API, guaranteeing accuracy. We discovered a lack of variability in the dislikes and trending columns, due to dislike stats being removed by YouTube and trending dates being a snapshot of when the data was captured by the author. Each country is separated by file, making analysis at country level more difficult. There are 3 missing channel titles and descriptions had a moderate amount of NA values that were determined to be “no description” rather than an error in collection. However, these are all relatively minor issues that can be easily fixed, the most pressing issues related to the duplicate records caused by a video trending in multiple countries. While this is great for country level analysis, this significantly skewed anything at the video level, as some videos would have 10 or 11 duplicate records. Perhaps an even bigger issue is that the duplicate videos would have different view, like, and comment counts. This is likely due to the author having collected them at different times, or in other words they could not collect all the countries at the same time. Due to this, the trending videos which are noted to be fast growing, would have higher view, like, and comment counts than the

same video in other countries. Note that this is not a view, like, and comment count for that country specifically, but a total for the video.

Pre-LLM Data Wrangling

Before our integration of LLMs, we did some data wrangling based on our own perceptions of what would be useful as well as to fix the major quality issues mentioned in the preliminary data quality assessment. First, due to each country being separated by file, it was immediately obvious we would need to combine these into a single dataset. To make this easy, we loaded each file and added a country column to them before appending them into a single data frame called YT_trending. Next, dislikes and trending_date columns were removed. The dislikes column was all 0's, due to YouTube removing the ability to see dislike data. Furthermore, the trending_date column was all the same date, making for no variability. Outside of this, we conducted some obvious feature engineering, such as creating columns for whether a video has a description or tags, as well as number of tags, description length, and video/channel title length. Outside of this, we decided that to fix the issue of the same video having different views, like, and comment counts across different countries, we used the maximum value for each. In other words, we took the maximum value for a video across the countries it trended in, which would be the latest snapshot of the counts for that video.

LLM implementation approach

We have implemented a local LLM solution that allows us to directly query from our R Markdown file. We utilized the Ellmer library alongside the Ollama tool for this task.

Ollama is a tool designed to allow you to run local LLMs rather than through a web interface or API. It gives access to models such as Googles' Gemma and Metas' Llama. For Ollama, you can download a model simple by running the command “ollama run [modelname]” in the terminal. For example, we used the model Gemma 3 12 billion parameter version, downloaded like this: “ollama run gemma3:12b”. Next, we use the ellmer library and create an object as follows: `chat <- chat_ollama(model = “gemma3:12b”, system_prompt = prompt)`. For our use case, we created a system prompt that gave a general overview of our data to the LLM, because we were unable to feed it the whole dataset due to the size. Our system prompt ended up being like the following:

```
You are a data analyst assistant.
Your task is to suggest useful visualizations and insights

Dataset Overview:
Rows: 178399
Columns: 16
Column names: video_id, title, channel_title, views, likes, publish_time, category_id, comments, channel_id, country, has_tags, num_of_tags, has_desc, desc_length, title_length, channel_title_length
Summary of numerical columns:
# A tibble: 1 x 36
  views_mean views_sd views_min views_max likes_mean likes_sd likes_min likes_max comments_mean comments_sd
  <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
1 1444214. 10814645.      0 1105386196 36071. 126959.      0 8968181 1253. 9773.
  comments_min comments_max has_tags_mean has_tags_sd has_tags_min has_tags_max num_of_tags_mean num_of_tags_sd
  <dbl>      <dbl>      <dbl>      <dbl>      <int>      <int>      <dbl>      <dbl>
1      0      3391804      0.742      0.437      0      1      13.0      12.3
  num_of_tags_min num_of_tags_max has_desc_mean has_desc_sd has_desc_min has_desc_max desc_length_mean
  <int>      <int>      <dbl>      <dbl>      <int>      <int>      <dbl>
1      0      125      0.875      0.331      0      1      810.
  desc_length_sd desc_length_min desc_length_max title_length_mean title_length_sd title_length_min
  <dbl>      <int>      <int>      <dbl>      <dbl>      <int>
1      954.      0      5000      52.5      23.8      1
  title_length_max channel_title_length_mean channel_title_length_sd channel_title_length_min
  <int>      <dbl>      <dbl>      <int>
1      105      11.0      5.73      1
  channel_title_length_max
  <int>
1      56
```

```
Summary of category_id:
# A tibble: 15 x 2
  category_id count
    <dbl> <int>
1         1  9342
2         2   759
3        10 44653
4        15   382
5        17  1923
6        19   181
7        20 76189
8        22 11570
9        23   894
10       24 28999
11       25  1393
12       26   862
13       27   439
14       28   790
15       29    23

Summary of countries:
# A tibble: 11 x 2
  country      count
    <chr>      <int>
1 Brazil    16200
2 Canada    16200
3 France    16200
4 Germany    16200
5 India     16199
6 Japan     16200
7 Mexico    16000
8 Russia    16200
9 South Korea 16200
10 United Kingdom 16400
11 United States 16400
```

Unique values for character columns (non chategorical)

```
# A tibble: 1 x 5
  video_id title channel_title channel_id country
    <int> <int>      <int>      <int> <int>
1  133188 128651      13077      13089    11
```

Some sample rows:

```
# A tibble: 5 x 16
  video_id title
    <chr>      <chr>
1 1tc8FEaVx5Y ГЕЙМПЛЕЙ МК 1: РЕАКЦИЯ НА ТРЕЙЛЕР ИГРОВОГО ПРОЦЕССА MORTAL KOMBAT 1
2 SwJ8u1g1lwM Fell For The Dead Sister Rage Bait | Dead Account
3 ZyF_JQY2Y1s Metallica Marching Band Competition: For Whom the Band Tolls!
4 obSutkLXF4E How to get Spongebob El Primo FOR FREE!
5 VPNFkn43DnA #Video - #Raushan Rohi - एक करोड़ - #रंगदारी #मगही गाना - Ek Karod - Rangdari Magahi Song 2025
  channel_title views likes publish_time category_id comments channel_id
    <chr>      <dbl> <dbl> <dtm>      <dbl>      <dbl> <chr>
1 Mikail      74074  3470 2023-06-09 11:19:05      20      440 UCGEsW1YNODzWqi647nG-lNg
2 Crunchyroll 11617   377 2026-01-29 14:00:46       1       14 UC6pGDc4bFGD1_36IKv3FnYg
3 Metallica   223875 12923 2023-04-10 14:00:31      10      618 UCbulh9WdLtEXiooRcYK7SWw
4 Brawl Stars 42979475 403679 2024-09-04 10:30:59      20     8661 UCooVYzDxdwTtGYAKcPmOgOw
5 Raushan Rohi Entertainment 3765430 60734 2025-03-31 05:19:41      10     2614 UCh6r40fbLs62rieVnODZkSw
  country has_tags num_of_tags has_desc desc_length title_length channel_title_length
    <chr>      <int>      <int>      <int>      <int>      <int>      <int>
1 Russia      1         23         1         945         67         6
2 Germany     1         6         1        1161         49        11
3 France      1         8         1        1849         61         9
4 Russia      1        18         1         282         39        11
5 India       1        15         1        1434         94        26
```

In other words, we fed summary statistics of our data so the LLM would have a general idea of what our dataset looks like and could answer our questions on it. After the object is created, to query the model we simply do `chat$chat("Some message.")`. With this, we asked our most important question to the LLM with the query: `chat$chat("Is there any necessary data cleaning, data integration, or data enrichment that needs to be done for this dataset?")`. The LLM responded with the following recommendations:

Okay, let's break down potential data cleaning, integration, and enrichment steps that would be beneficial for this dataset. Given the description and summary information, here's a prioritized list with explanations:

****1. Data Cleaning (Highest Priority)****

- * **publish_time` Data Type Correction:** The summary mentions `publish_time` is a `dtm` (date-time) object. It's *critical* to verify this. Often, date/time columns are imported as strings. If it's a string, you *must* convert it to a proper date/time format for analysis. Incorrect formats will wreck time-series analysis or any date-based calculations.
- * **Missing Values:** While summaries don't explicitly mention missing values, it's a standard practice to check *every* numerical column (`views`, `likes`, `comments`, `num_of_tags`, `desc_length`, `title_length`, `channel_title_length`) for `NA` values. Handle them appropriately (imputation with mean/median/mode, or removal if the missingness is substantial and biased).
- * **Outlier Handling for `views`:** The `views_sd` is *huge* compared to `views_mean`. This indicates a very wide distribution and likely contains outliers. Extreme values in `views` can skew any analysis. Consider:
 - * Visual inspection (histograms, boxplots) to identify outliers.
 - * Winsorizing or trimming extreme values to reduce their impact. (Be careful - might be legitimate viral videos!)
- * **category_id` Consistency:** Look for any unexpected or invalid `category_id` values. Occasionally there might be errors in data entry (e.g., a value that doesn't belong to the established categories).
- * **String Column Cleaning (Title, Channel Title, Description):**
 - * **Whitespace:** Trim leading and trailing whitespace from `title`, `channel_title`, and `desc`.
 - * **Character Encoding:** Ensure consistent character encoding (e.g., UTF-8) to handle special characters correctly. Problematic encodings can lead to errors in text analysis.
 - * **HTML/Special Characters:** The descriptions might contain HTML tags or other special characters that need to be removed or replaced.
- * **country` Consistency:** While the summary shows a list of countries, ensure consistency in country names (e.g., "United States" vs. "USA"). Standardize to a consistent format.

****2. Data Integration (Medium Priority)****

* ****External Data (Channel Stats):**** Ideally, integrate with external data sources to enrich the analysis. For example:

* ****Channel Statistics:**** Combine this dataset with channel-level data (subscriber counts, average views per video, etc.) to understand channel performance drivers. This would likely involve using `channel\_id` as a join key.

* ****Trending Data:**** Integrate with external trending data (e.g., Google Trends) to see how video popularity aligns with broader search interest.

* ****Geographic Data:**** Integrate with geographic data to understand regional trends in video consumption.

****3. Data Enrichment (Lower Priority, but Valuable)****

* ****Text Analysis (Title, Description, Channel Title):****

* ****Topic Modeling:**** Using techniques like Latent Dirichlet Allocation (LDA), apply topic modeling to the `title` and `description` columns to identify common themes.

* ****Sentiment Analysis:**** Analyze the sentiment (positive, negative, neutral) expressed in the `title` and `description`.

* ****Keyword Extraction:**** Extract relevant keywords from the `title` and `description` to understand video content.

* ****Derived Features:****

* ****Time-Based Features:**** Extract features from `publish\_time`, like:

* Day of the week

* Month

* Year

* Hour of day

* ****Ratio Features:**** Create ratios like:

* `likes / views` (Engagement Rate)

* `comments / views` (Interaction Rate)

* `desc\_length / title\_length` (Description Length Relative to Title)

* ****Video Age:**** Calculate the age of the video from `publish\_time` to the present day.

****Prioritization Rationale:****

1. ****Cleaning**** is essential for accurate and reliable analysis.

2. ****Integration**** can provide valuable context and insights.

3. ****Enrichment**** adds depth and nuance but may not be critical for initial exploration.

To fully determine the necessary steps, a sample of the data would be helpful to inspect the actual data quality and format.

For the most part, these are solid recommendations from the LLM and helped us with our data. There were a few small unnecessary ones, like checking date time, as the LLM should have already knew this information is accurate due to the system prompt. Nonetheless, we followed some of the recommendations, particularly related to outlier handling, string column cleaning, time-based feature creation, ratio features, and video age. Outside of this, we used the LLM to recommend visualizations for research questions

we wanted to answer. The following are the queries and responses for our questions on visualizations:

```
chat$chat("Please recommend an appropriate visualization with a short explanation as to why it would be appropriate to answer the following question: Do views translate into engagement and interaction? (Does more views cause an increase in likes and comments?)")
```

****Visualization:** Scatter Plot (Views vs. Likes & Views vs. Comments)****

****Explanation:**** A scatter plot is the most direct and effective visualization to address this question. Each dot on the plot represents a single video. By plotting "Views" on the x-axis and "Likes" (or "Comments") on the y-axis, we can visually assess if there's a positive relationship: do videos with more views generally have more likes/comments? A rising trendline would suggest that views do indeed translate into engagement. It's a simple, easily understandable way to observe potential correlations between these key metrics.

It's the baseline for exploring the question because it directly displays the relationship between the variables we're interested in.

```
chat$chat("Please recommend an appropriate visualization with a short explanation as to why it would be appropriate to answer the following question: Does category effect engagement (likes), interaction(comments), and overall viewership?")
```

****Visualization:** Box Plots (One for Each Category Metric)****

****Explanation:**** Box plots are ideal for comparing the distribution of numerical data (likes, comments, views) across different categories (the video categories). We'd create three separate box plots:

- * ****Category vs. Likes:**** Shows the distribution of likes for each category.
- * ****Category vs. Comments:**** Shows the distribution of comments for each category.
- * ****Category vs. Views:**** Shows the distribution of views for each category.

Each box plot will display the median, quartiles, and potential outliers for each category. This allows us to visually compare the central tendency and spread of likes, comments, and views across the different video categories. We can quickly see if certain categories consistently have higher median likes, more comments, or greater viewership compared to others.

Box plots are particularly good because they handle multiple categories in a clear way, highlight differences in data spread, and readily point out any outliers that might be skewing the overall picture.

```
chat$chat("Please recommend an appropriate visualization with a short explanation as to why it would be appropriate to answer the following question: Do newer videos tend to be more engaging and interactive or less?")
```

****Visualization:** Line Chart (Engagement/Interaction Metrics vs. Time)****

****Explanation:**** A line chart is perfect for visualizing trends over time. We'll create separate line charts for "Likes," "Comments," and potentially "Views" (depending on whether you suspect viewership also changes with video age).

- * ****X-axis:**** Time (publish_time). We'll need to convert this into a suitable time scale - perhaps days, weeks, or months since publish.
- * ****Y-axis:**** The engagement/interaction metric we're analyzing (Likes, Comments, Views).

The line chart will then show how these metrics change as videos age. A rising line indicates that engagement/interaction *increases* with video age, while a declining line suggests it *decreases*. This directly illustrates the trend of engagement over time.

****Why it's appropriate:**** Line charts are specifically designed to show how a variable changes across a continuous variable like time, making it ideal for spotting trends.

```
chat$chat("Please recommend an appropriate visualization with a short explanation as to why it would be appropriate to answer the following question: Does having tags and a description increase viewership? Does length of description/title and number of tags matter?")
```

****Visualization: Scatter Plots & Correlation Matrix****

****Explanation:**** This question requires exploring multiple correlations, so a combination of visualizations is best.

1. ****Scatter Plots (Initial Exploration):****

- * **`Has\_Tags` vs. Views`:** Scatter plot with 'Has_Tags' (likely a binary variable: 0 = No Tags, 1 = Yes) on the x-axis and 'Views' on the y-axis. This will show if videos with tags generally have higher viewership.

- * **`Description\_Length` vs. Views`:** Scatter plot with 'Description_Length' (number of characters/words in the description) on the x-axis and 'Views' on the y-axis.

- * **`Title\_Length` vs. Views`:** Scatter plot with 'Title_Length' on the x-axis and 'Views' on the y-axis.

- * **`Num\_Tags` vs. Views`:** Scatter plot with 'Num_Tags' (number of tags) on the x-axis and 'Views' on the y-axis.

2. ****Correlation Matrix (Summary):****

- * Create a correlation matrix including 'Views', 'Has_Tags', 'Description_Length', 'Title_Length', and 'Num_Tags'. Visualize this as a heatmap.

****Why this combination is appropriate:****

- * ****Scatter Plots:**** Reveal potential relationships (positive, negative, or no correlation) between the presence/length of descriptions/tags and viewership.

- * ****Correlation Matrix:**** Provides a quantitative summary of the correlations between all these variables. It's a quick way to see which factors are most strongly associated with viewership.

By analyzing these visualizations together, we can get a comprehensive understanding of whether having tags and descriptions, and their lengths/numbers, influence video views.

```
chat$chat("Please recommend an appropriate visualization with a short explanation as to why it would be appropriate to answer the following question: Does time of day/week affect viewer numbers?")
```

****Visualization: Heatmap (Time of Day/Week vs. Views)****

****Explanation:**** A heatmap is ideal for visualizing the relationship between two categorical variables—in this case, time of day and day of the week—and a numerical variable (views).

- * ****Rows:**** Day of the Week (Monday, Tuesday, Wednesday, etc.)

- * ****Columns:**** Time of Day (e.g., broken down into hourly or 2-hour blocks: 6 AM - 8 AM, 8 AM - 10 AM, etc.)

- * ****Cell Color:**** Represents the average or total number of views for videos published during that specific time slot and day of the week. Darker colors could indicate higher view counts.

This allows you to quickly spot patterns: Are videos published on weekends evenings consistently more viewed than those published on weekday mornings? It elegantly combines two categorical variables to reveal when viewer numbers are at their peak.

Heatmaps are excellent for identifying patterns in data where you want to see the combination of two categorical factors and their impact on a third numerical variable.

```
chat$chat("Please recommend an appropriate visualization with a short explanation as to why it would be appropriate to answer the following question: Do certain channels dominate the trending charts, or is there significant variance?")
```

Visualization: Bar Chart (Channel Viewership/Popularity)

Explanation: To assess whether certain channels dominate the trending charts or if there's significant variance, a bar chart is the most straightforward and effective visualization.

- * **X-axis:** Channel Titles (or Channel IDs, if more compact representation is needed)
- * **Y-axis:** A measure of channel popularity/trending status. This could be:
 - * **Total Views:** Displaying the total views for each channel.
 - * **Number of Trending Videos:** Counting how often videos from each channel appear on trending charts (if that data is available - otherwise total views is a good proxy).
 - * **Average Views per Video:** Showing the average views across all videos published by each channel.

The bar chart will clearly show which channels have the highest viewership/trending presence. If a few channels have significantly taller bars compared to the rest, it indicates channel dominance. A more even distribution of bar heights suggests greater variance and less channel domination.

Bar charts are perfect for comparing discrete entities (in this case, channels) based on a quantifiable metric, allowing a clear visual assessment of dominance or variance.

```
chat$chat("Please recommend an appropriate visualization with a short explanation as to why it would be appropriate to answer the following question: Do channels with a lot of trending videos do anything different? For example, perhaps channels with more trending videos are more likely to have a description and tags. Perhaps channels with more trending videos tend to make longer descriptions or put more tags (this is speculation, what we want to figure out).")
```

Visualization: Scatter Plots & Grouped Bar Charts (Comparative Analysis)

Explanation: This question requires a comparative look at channel characteristics based on their trending video activity. A combination of scatter plots and grouped bar charts will be most informative.

1. **Scatter Plots (Exploring Correlations):**
 - * **Trending Videos vs. Has_Tags:** Scatter plot with "Trending Videos" (number of trending videos a channel has) on the x-axis and "Has_Tags" (binary: 0/1) on the y-axis. This checks if channels with more trending videos are more likely to use tags.
 - * **Trending Videos vs. Description_Length:** Scatter plot with "Trending Videos" on x and "Description_Length" on y. Tests if trending channels have longer description lengths.
 - * **Trending Videos vs. Num_Tags:** Scatter plot with "Trending Videos" on x and "Num_Tags" on y. Tests if trending channels use more tags.
2. **Grouped Bar Charts (Comparing Averages):**
 - * Create two groups: Channels with a *low* number of trending videos and channels with a *high* number of trending videos. Define "low" and "high" based on quantiles or a reasonable split of the data.
 - * For each group, create grouped bar charts comparing:
 - * Average "Has_Tags"
 - * Average "Description_Length"
 - * Average "Num_Tags"

Why this combination is appropriate:

- * **Scatter Plots:** Show if there's a *correlation* (even if not causation) between trending video counts and description/tag usage.
- * **Grouped Bar Charts:** Provide a direct comparison of *average* values of these characteristics between high-trending and low-trending channels. They'll highlight if trending channels, on average, are doing things differently regarding descriptions and tags.

This layered approach allows for a comprehensive exploration of the hypothesis, revealing both potential correlations and average differences between channel groups.

Once again, these are solid recommendations. We implemented these visualizations almost exactly, doing minor editing where it made sense.

LLM Integration Discussion

To conclude our LLM integration approach, we believe this approach was the most suitable for our needs. LLMs have the ability to hallucinate, which can severely compromise the accuracy and completeness of a project, making full automation difficult and dangerous. Therefore, we simply opted for this “recommendations” approach instead. For the most part, the LLM helped reduce the time spent deciding what data wrangling and analysis we should do. The work, however, was done by us, allowing for human validation and correction on its methods. However, if I were to criticize this approach, I would say that overall, a local LLM solution is simply not good at the level of model we can run on basic consumer hardware. Perhaps with powerful enough hardware, we could run a much more powerful model capable of holding more information about our data, which would allow it to be less “basic” and more nuanced to our specific needs. In the near future, either using an API or web interface based LLM (such as ChatGPT) gives easier access to more capable models.

Discussion

First, to begin with here are some of our summary statistics:

video_id	trending_date	title	channel_title
Length:178399	Length:178399	Length:178399	Length:178399
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

views	likes	dislikes
Min. :0.000e+00	Min. : 0	Min. :0
1st Qu.:4.602e+04	1st Qu.: 837	1st Qu.:0
Median :2.313e+05	Median : 5199	Median :0
Mean :1.445e+06	Mean : 36086	Mean :0
3rd Qu.:8.326e+05	3rd Qu.: 22765	3rd Qu.:0
Max. :1.105e+09	Max. :8968181	Max. :0

publish_time	category_id	tags
Min. :2006-08-25 14:14:26	Min. : 1.0	Length:178399
1st Qu.:2024-12-02 20:13:25	1st Qu.:10.0	Class :character
Median :2025-08-21 19:15:06	Median :20.0	Mode :character
Mean :2025-01-13 19:50:45	Mean :17.3	
3rd Qu.:2026-01-06 08:00:25	3rd Qu.:22.0	
Max. :2026-02-26 16:00:51	Max. :29.0	

comments	channel_id	description	country
Min. : 0	Length:178399	Length:178399	Length:178399
1st Qu.: 28	Class :character	Class :character	Class :character
Median : 187	Mode :character	Mode :character	Mode :character
Mean : 1253			
3rd Qu.: 752			
Max. :3391804			

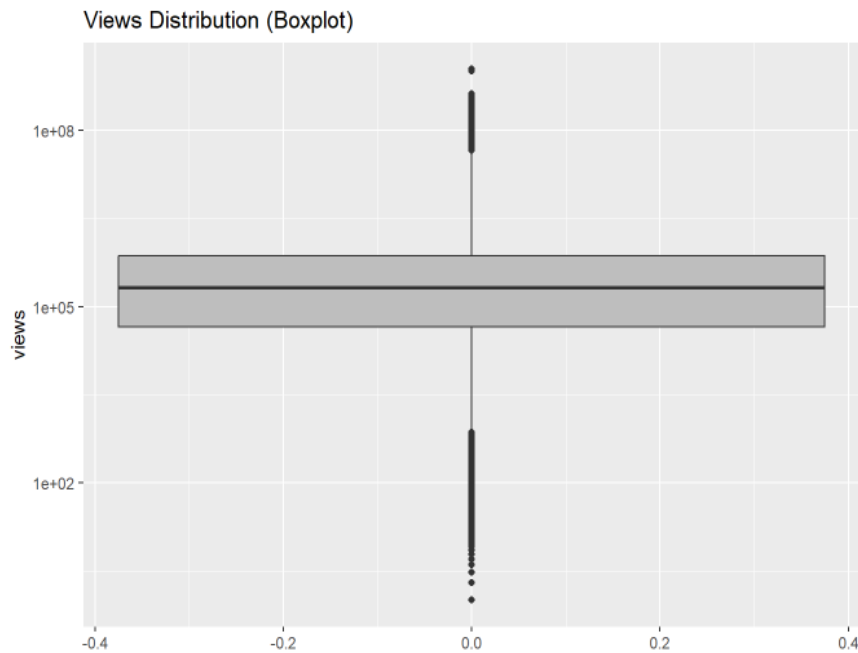
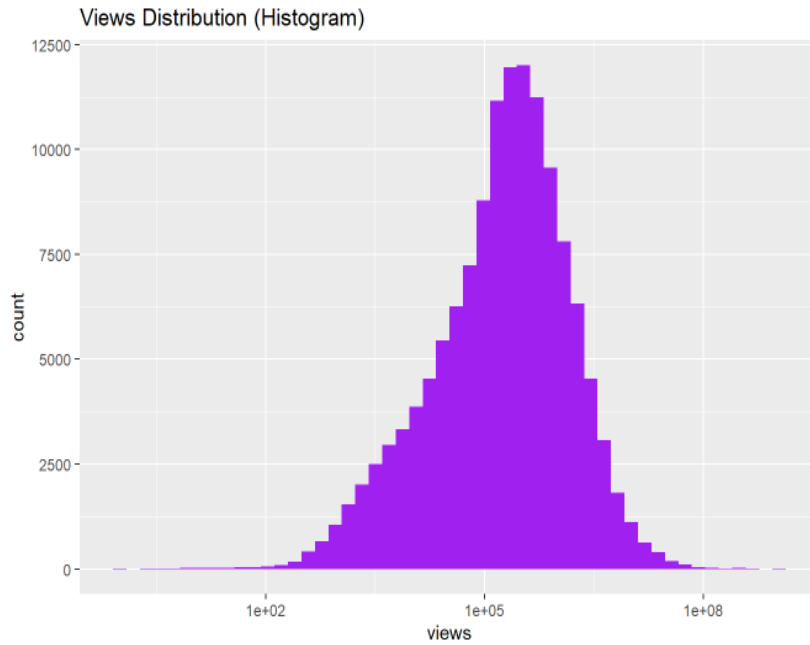
video_id	title	channel_title	views	likes
0	0	3	0	0

publish_time	category_id	tags	comments	channel_id
0	0	0	0	0

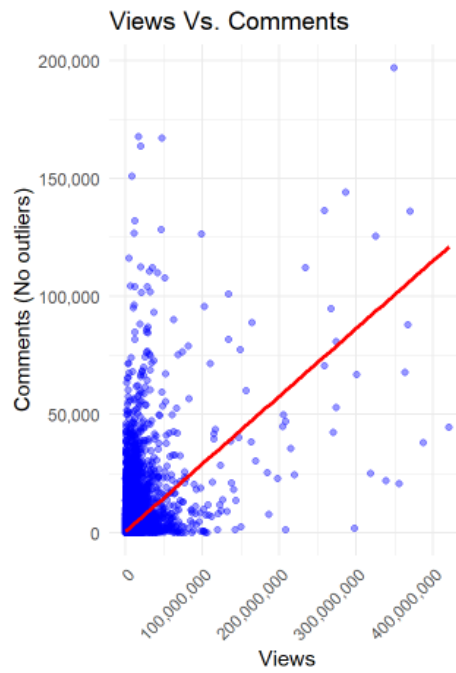
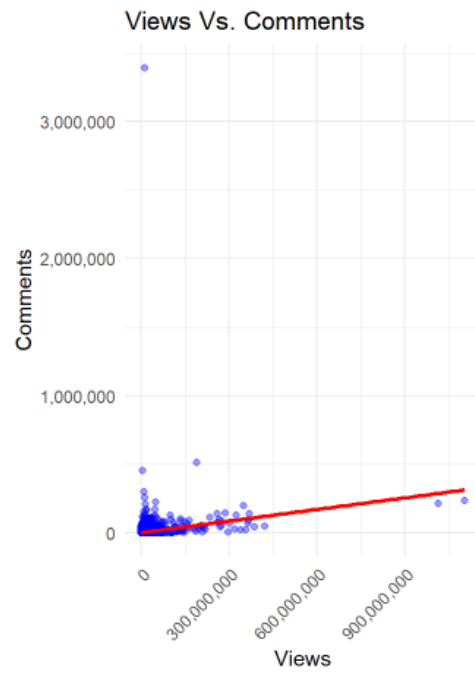
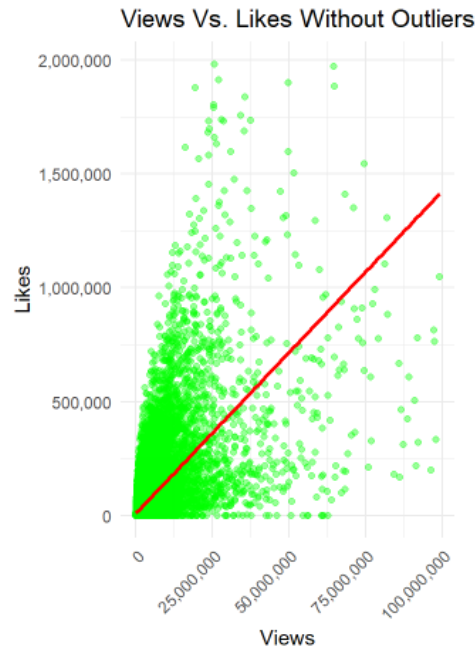
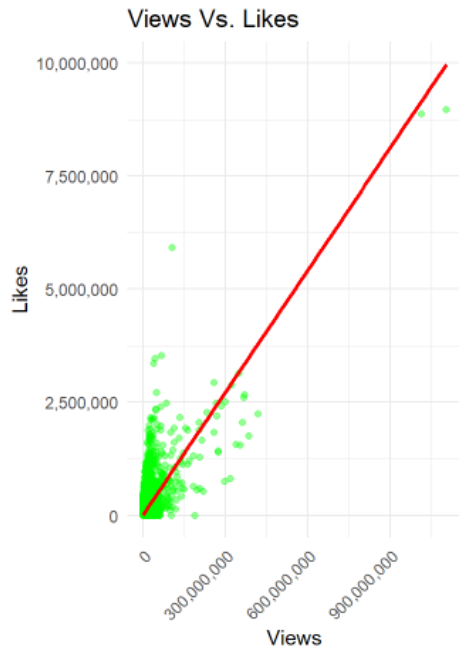
description	country
22325	0

For our summary statistics above, we can see clearly that our views, like, and comments all have significantly higher max values compared to mean/median, indicating the presence of outliers. Therefore, outliers were removed for subsequent visuals. As previously discussed in our preliminary data quality assessment, dislikes are all 0s. We also talked about the three channel_title NA values and the 22,325 NA description values, which are determined to be “no description” rather than missing values.

Here are our visuals we made:

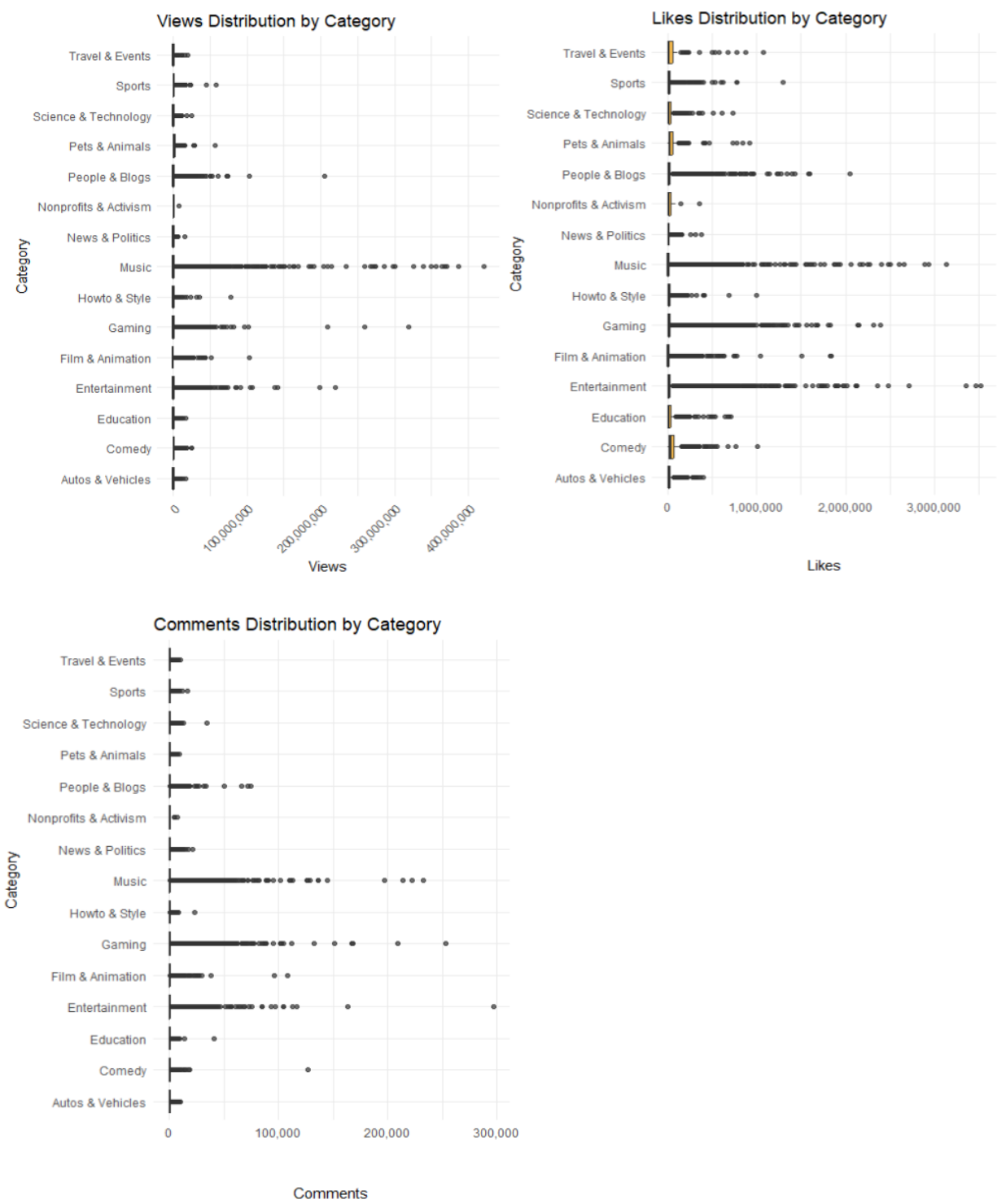


The histogram and boxplot above show us the views distribution among all videos in the dataset. For the most part, videos tend to go around the 100,000 views mark, with some extreme outliers on both ends.

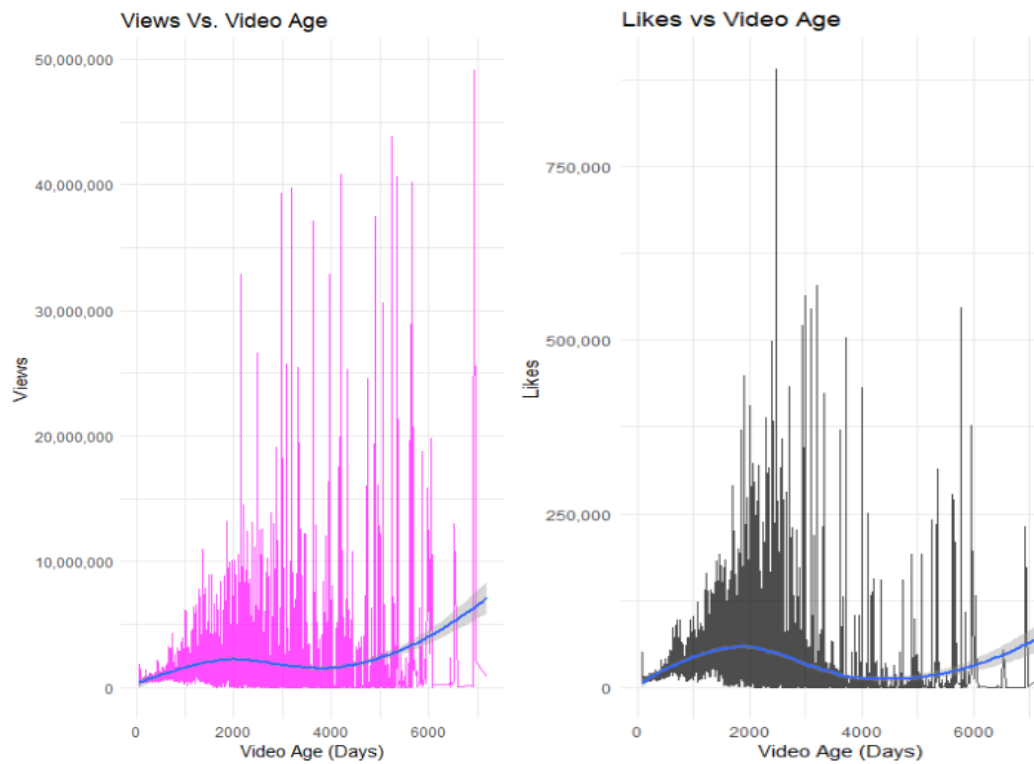


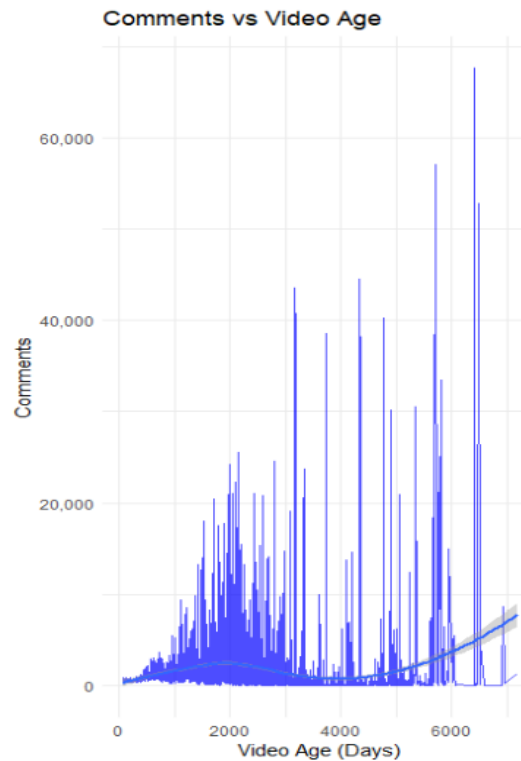
These graphs answer the question of whether more views lead to more comments and likes respectively. The graphs on the right are without any outliers. As expected, the

graphs show that as the number of views increases, the number of comments and likes tend to increase as well.

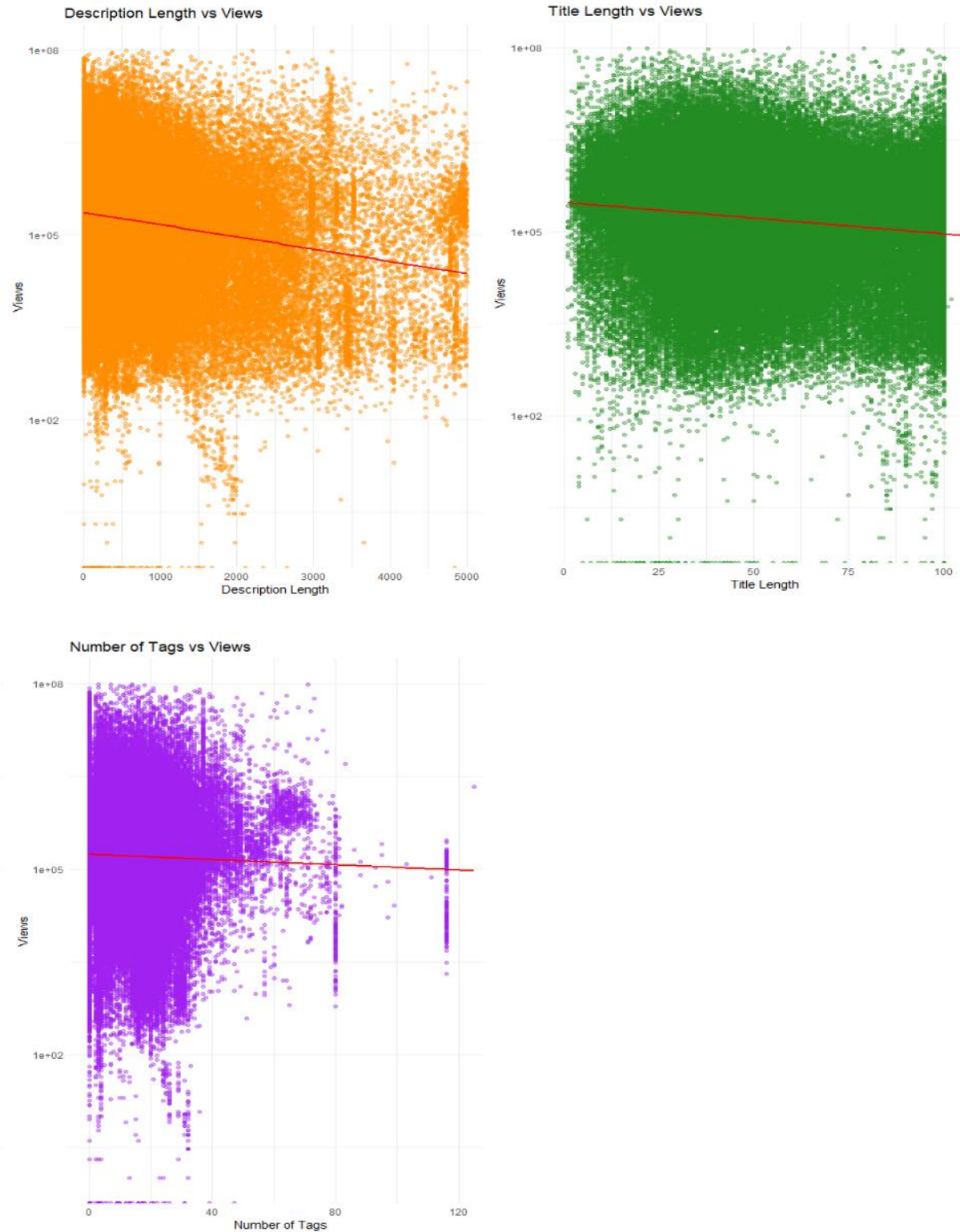


These graphs answer the question about whether categories affect the number of likes, comments, and views respectively. The highest categories include music, gaming, and entertainment. It is very clear that videos in these categories tend to perform significantly better across all metrics than Nonprofits & Activism, for example.



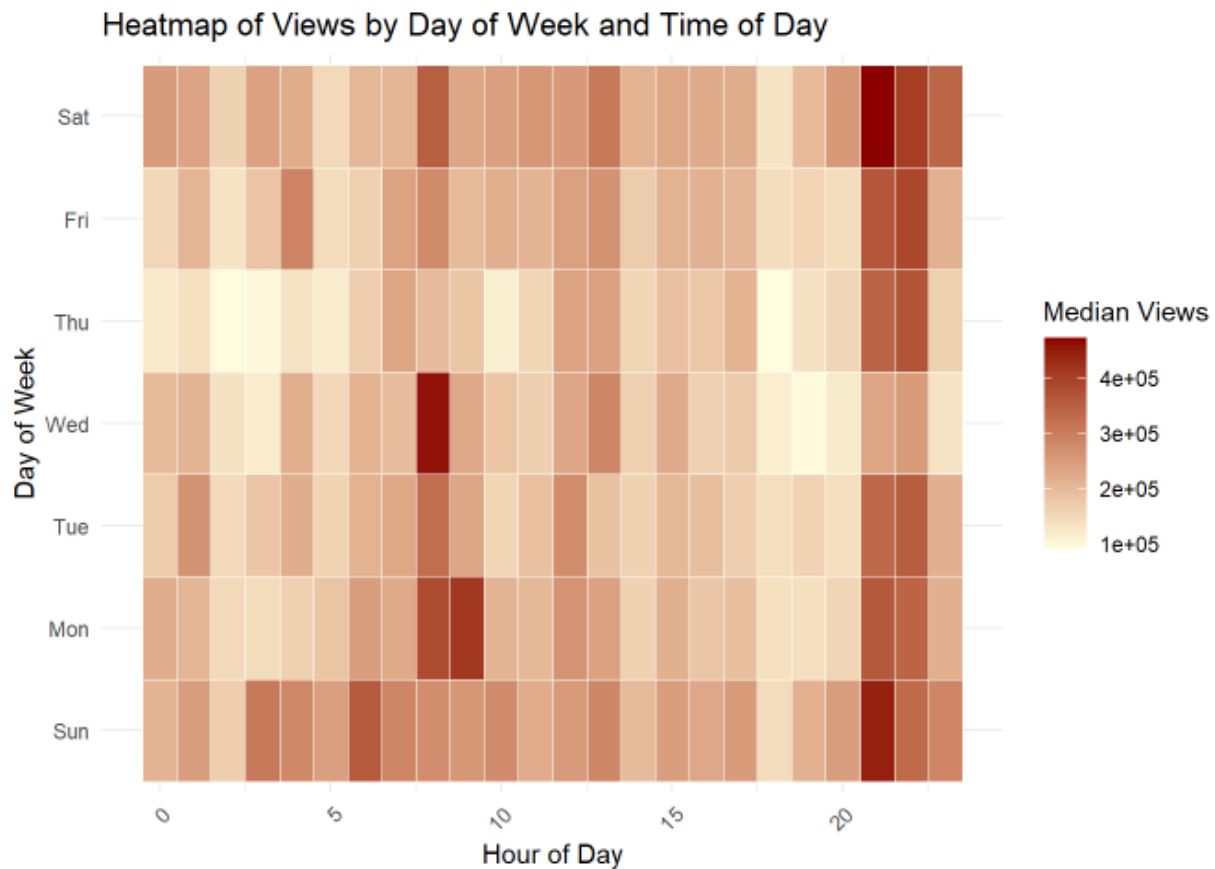


These graphs answer the question of whether newer videos tend to be more engaging or interactive than older videos. For the most part, videos tend to have a higher view, like, and comment count the older they get. However, there is a noticeable bump around the 2,000-day mark. To investigate, 2,000 days ago would put us in the year 2020. Obviously, this is the year of the COVID-19 pandemic, which coincided with stay-at-home orders, quarantines, and more issues that required people to stay at home for longer. This, alongside a digital world that constantly grows, is likely the cause of the bump during that period.



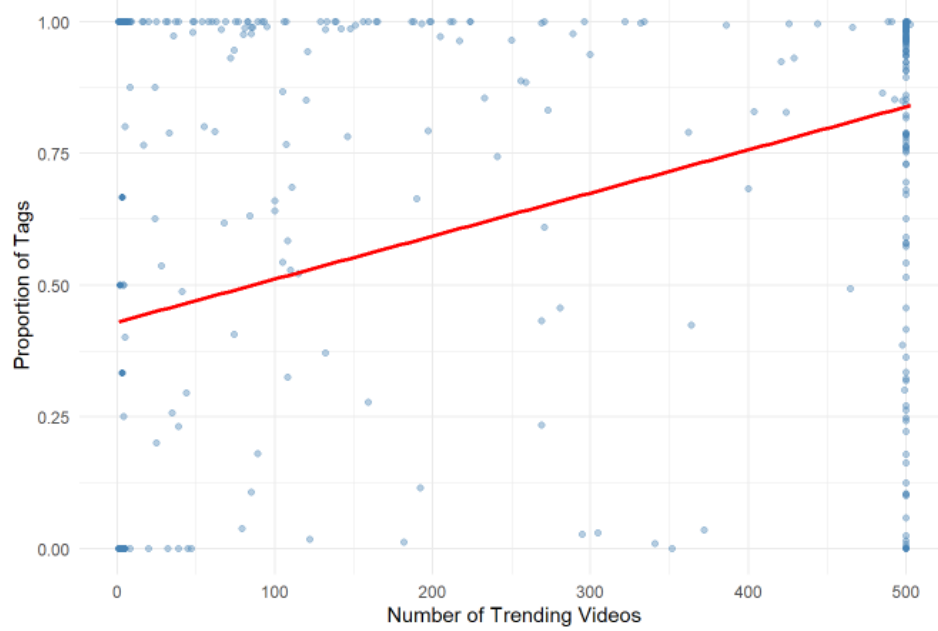
These graphs answer the question of whether tag amount, description, and title length affect views. As the graphs show, there is a slight negative impact on the views as

the description length, title length, and number of tags increase. Perhaps this is because length does not equal quality, and the quality of these things far outweigh the length.

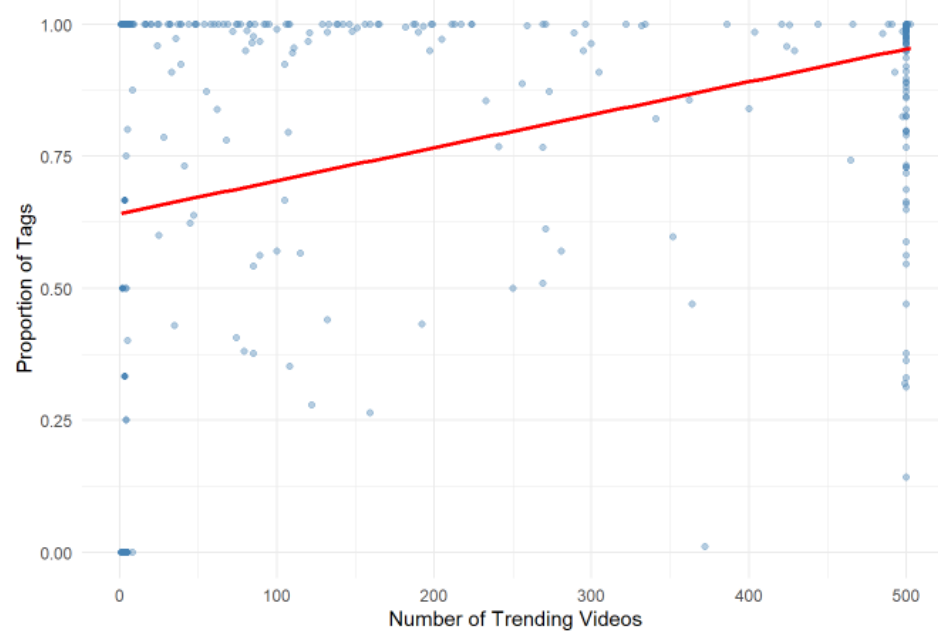


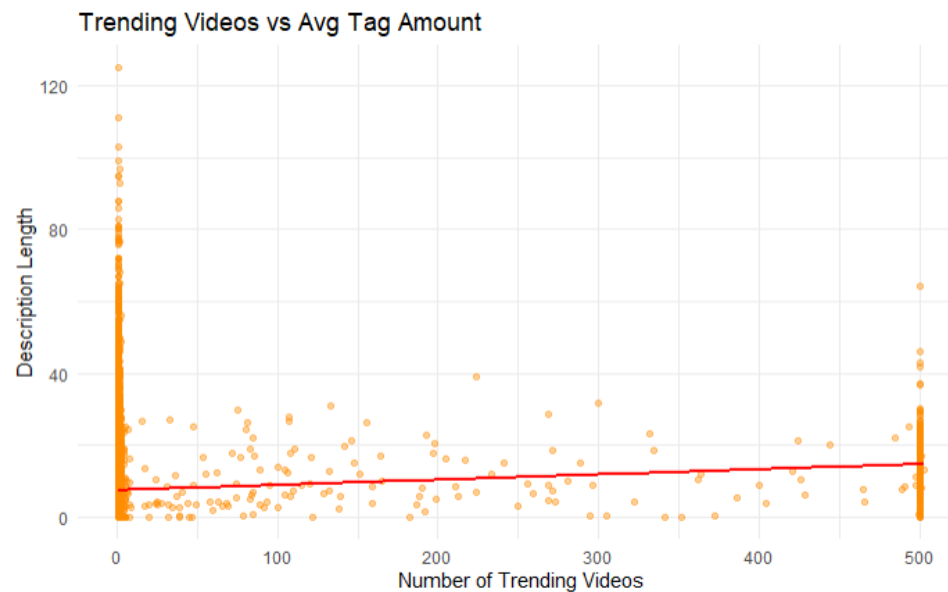
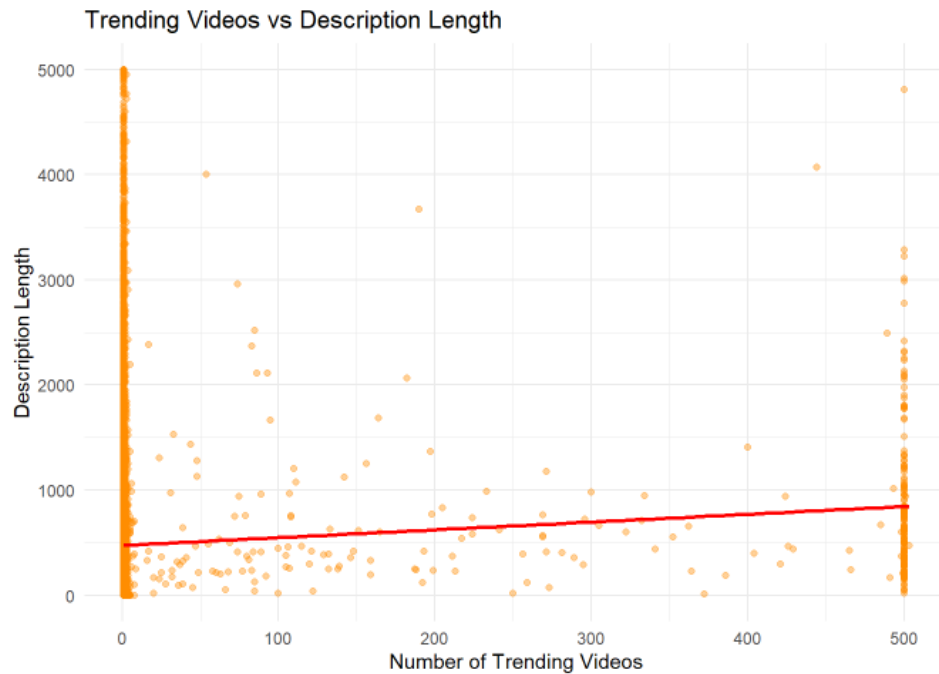
This graph answers the question of whether the time of day and day of the week affect views. This is evident during the 21st and 22nd hours of the day as these columns tend to have higher median values than other columns. An initial guess for why this may be is that around 9:00pm (21 hours) is when most people are finally able to relax after a long day of work. Meaning they get home, shower, clean, eat food, etc and then that tends to be when they are finally able to relax. Then, at around 11:00pm (23 hours) people start to wind down and go to sleep.

Trending Videos vs Proportion of Tags

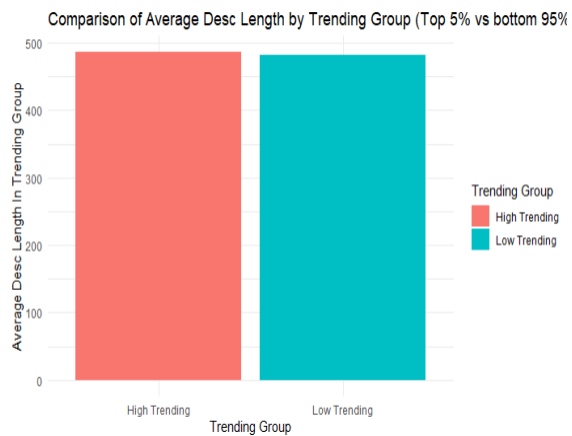
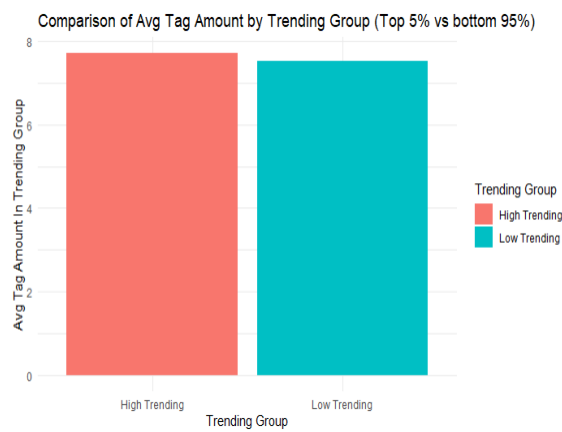
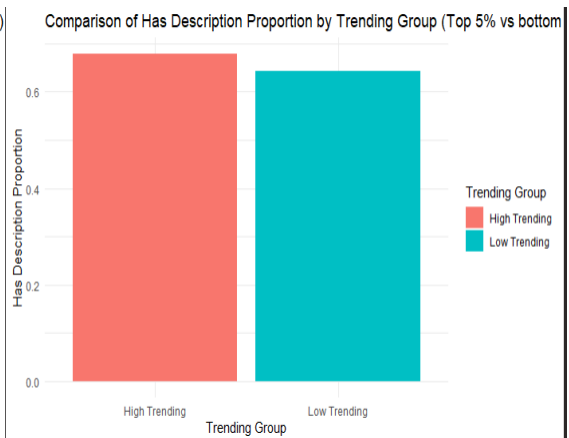
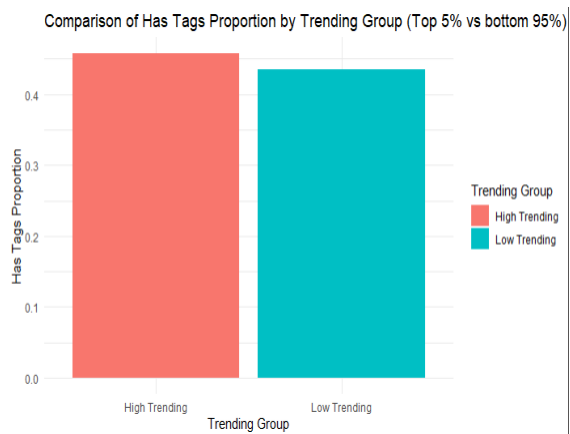


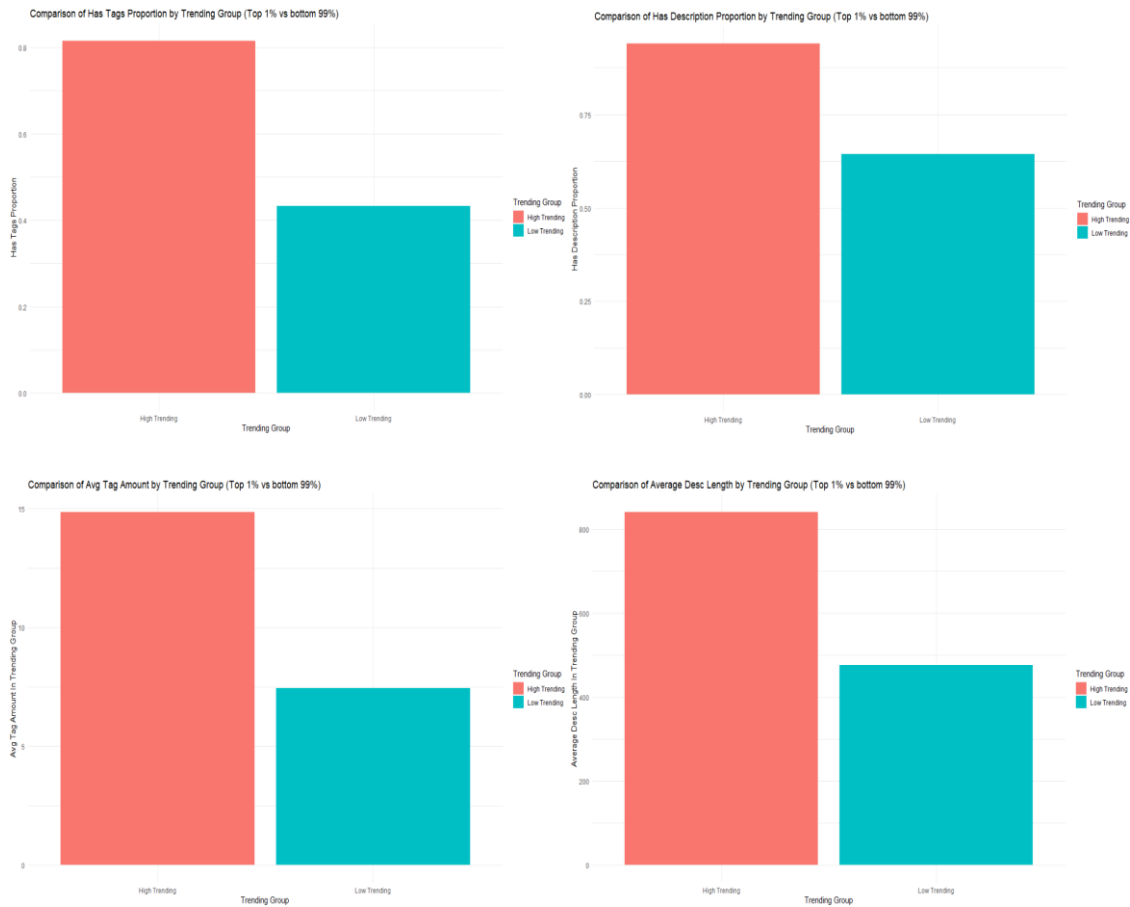
Trending Videos vs Proportion of Desc





These graphs show whether the number of trending videos on a channel affects the tag proportion, description proportion, description length, and tag count respectively. It tends to be that channels with more trending videos have a higher portion of their videos with a description and tags. These channels also tend to have longer descriptions and more tags, but these trendlines are only slightly positive.





These 8 graphs group together channels by whether they are “high trending” (high number of trending videos) or “low trending” (low number of trending videos). For the first four graphs, high trending is top 5% and low trending is bottom 95%, while the last four graphs are top 1% and bottom 99% respectively. While the first four graphs do not show much difference between top 5% and bottom 95%, the top 1% tend to have a significantly higher average description length, tag amount, as well as tag and description proportion. For this, we conclude that including a tag and description is a benefit towards trending more often. It is also beneficial to make longer descriptions and more tags, but this likely tapers off as the quality of the tags and descriptions likely make a stronger difference.

Quality is difficult to quantify, so research needs to be done about what makes quality tags and descriptions.

Conclusion and Future Work

In this paper, we have discussed the YouTube trending video statistics using the dataset “YouTube Trending Videos Stats 2026” by Shul Ilya. Our paper mainly looked at the effects of tags and descriptions, and we found that videos with higher stats tend to have descriptions/tags more often, as well as longer descriptions and more tags. Furthermore, we discussed the usage of large language models in data wrangling and analysis, as well as implemented a local LLM solution to assist us with our tasks. However, this project is not perfect and can implement multiple improvements to make it more complete and give more accurate conclusions. First, the LLM integration could be improved by deploying a more powerful model. A more powerful model would in theory hallucinate less, give more accurate and “niche” recommendations (less basic), catch issues, and have a higher memory allowing for a bigger system prompt (more specific recommendations to our tasks). However, this would be difficult on the local LLM level due to hardware restrictions, so a web interface (ChatGPT) or API solution would be needed, at least in the near future. Further improvements on the analysis and data side mainly include incorporating different types of data rather than just more of the same. For example, a true channel statistics dataset (subscriber counts, view/like/comment counts, averages, medians and more) would be a massive boost compared to just making a trending videos level for channel

stats. Also, comparing high performance, non-trending videos could help improve overall analysis on what makes a video trend.

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